

Exact values



- 1 Consider the equation $\sin \theta = -\frac{1}{2}$. State the number of solutions for θ , in the domain $0 < \theta < \frac{\pi}{2}$.
- 2 For each of the following equations, find the exact value of acute angle x , in the domain $0 \leq x \leq \frac{\pi}{2}$:
- a $\tan x = 1$ b $\sin x = \frac{\sqrt{3}}{2}$ c $\cos x = \frac{1}{\sqrt{2}}$ d $2 \tan x = \frac{2\sqrt{3}}{3}$
- 3 For each of the following, find the exact values of x , where $-\pi < x < \pi$:
- a $\tan x = \sqrt{3}$ b $\tan x = \frac{1}{\sqrt{3}}$
- c $\sin 2x = \frac{1}{2}$ d $\tan \left(x - \frac{2\pi}{3} \right) = 0$
- 4 For each of the following, find the exact values of θ , in the given domain:
- a $\tan \theta = 0$, where $-\pi \leq \theta \leq \pi$ b $\sin \theta = 0$, where $-4\pi \leq \theta \leq 4\pi$
- c $12 \cos \theta - 6\sqrt{3} = 0$, where $-\pi < \theta < \pi$ d $4 \sin \theta - 3 = -5$, where $-2\pi \leq \theta \leq 2\pi$
- e $\cos^2 \left(\frac{\theta}{2} \right) - 1 = 0$, where $-2\pi \leq \theta \leq 2\pi$ f $4 \sin^2 (\theta) = 1$, where $-4\pi < \theta < 4\pi$
- 5 For each of the following, find the exact values of x , over the domain $[0, 2\pi)$:
- a $\sin x = \frac{1}{2}$ b $\cos x = -\frac{\sqrt{3}}{2}$
- c $6 \cos x + 2 = -1$ d $6 \cos x - 3\sqrt{2} = 0$
- e $2 \tan x + 3 = 5$ f $2 \sin x + 2 = 1$
- g $\cos x \tan x = \cos x$ h $\tan^2 (x) + 2 \tan x + 1 = 0$
- i $2 \sin 3x - \sqrt{2} = 0$ j $\sin \left(\frac{x}{2} \right) = \frac{\sqrt{3}}{2}$
- k $2 \sin \left(\frac{x}{2} \right) = \sqrt{3}$ l $\sin \left(\frac{x}{2} \right) = -\cos \left(\frac{x}{2} \right)$
- m $\cos \left(\frac{x}{2} \right) = 1 - \cos \left(\frac{x}{2} \right)$ n $\sin \left(\frac{x}{2} \right) = 1 - \sin \left(\frac{x}{2} \right)$
- o $\cos^2 \left(\frac{x}{2} \right) - 1 = 0$ p $\sin^2 \left(\frac{x}{2} \right) - 1 = 0$
- q $36(1 - \cos x)(1 + \cos x) = 27$

6 For each of the following, find the exact value of x , over the domain $0 \leq x \leq 2\pi$:

a $\sin x = 1$

c $\sin x = \frac{1}{2}$

e $\cos x = -\frac{1}{2}$

g $10 \sin x - 5\sqrt{3} = 0$

i $2\sqrt{3} \tan x - 2 = 0$

k $\tan 3x = -\frac{1}{\sqrt{3}}$

m $\cos\left(x + \frac{\pi}{6}\right) = -\frac{1}{2}$

o $\sin\left(x + \frac{\pi}{3}\right) = \frac{1}{2}$

q $\cos\left(x + \frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$

b $\cos x = -1$

d $\sin x = -\frac{1}{\sqrt{2}}$

f $\tan x = -\frac{1}{\sqrt{3}}$

h $4 \cos x - 2 = 0$

j $\sin 2x = \frac{1}{\sqrt{2}}$

l $\sin\left(x + \frac{\pi}{6}\right) = \frac{1}{\sqrt{2}}$

n $\sin x = -\cos x$

p $\sqrt{3} \tan\left(\frac{x}{2}\right) = -3$

r $\sin\left(x - \frac{4\pi}{3}\right) = \frac{1}{2}$

7 Find the exact solutions of the following equations where $0 \leq \theta \leq 2\pi$:

a $\cos^2 \theta - 5 \cos \theta + 4 = 0$

c $\tan^2 \theta + \sqrt{3} \tan \theta = 0$

b $2 \sin^2 \theta + \sin \theta - 1 = 0$

d $\sin^2 \theta - \cos^2 \theta = 0$

Other radian values

8 Consider the equation $\cos \theta = 0.42$. State the number of solutions for θ over the domain $0 < \theta < 2\pi$.

9 For each of the following equations:

i Find the acute angle θ in radians that solves the equation, correct to two decimal places.

ii Hence find all solutions to θ over the domain $[0, 2\pi]$, correct to two decimal places.

a $\sin \theta = 0.3420$

b $\cos \theta = 0.9063$

c $\tan \theta = 0.8340$

d $\cos \theta = 0.2345$

e $\sin \theta = 0.9921$

f $\tan \theta = 0.7743$

g $\cos \theta = 0.67$

h $\sin \theta = 0.091$

i $\cos 2\theta = 0.9$

10 Jessica finds the acute angle solution to the equation $\sin x = k$, is $x = p$.

Find another solution for the equation over the domain $0 < x < 2\pi$, in terms of p .

- 11 Toby finds the acute angle solution to the equation $\cos x = m$, is $x = n$.
Find another solution for the equation over the domain $0 < x < 2\pi$, in terms of n .
- 12 Sarah finds the acute angle solution to the equation $\tan x = h$, is $x = t$.
Find another solution for the equation over the domain $0 < x < 2\pi$, in terms of t .